

TULCEA CATALOI, Romania

WMO: 153350

Lat: 45.1906N

Lon: 28.8242E

Elev: 18

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	15.7	5.3	7.2	13.5	10.5	9.4	18.1	27.4	31.3	22.9	34.6	7.4	0	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	91.6	71.7	88.9	70.7	86.3	69.8	75.2	85.3	73.6	83.2	72.1	81.9	6.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	119.9	79.6	70.6	112.8	78.2	69.0	106.5	76.8	38.7	85.6	37.2	83.1	35.8	81.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	18.3	15.8	3.8	96.2	4.8	3.7	0.3	98.9	-2.5	101.0	-5.2	103.1	-8.7	105.9
			3.3	78.9	4.7	2.7	-0.1	80.8	-2.9	82.4	-5.5	83.9	-8.9	85.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	31.9	35.8	43.3	53.1	63.2	70.9	74.6	73.9	64.8	54.2	44.7	35.5	(6)
	DBStd	16.53	9.05	9.22	7.86	6.30	5.87	5.20	4.22	5.23	6.07	7.33	8.37	8.45	(7)
	HDD50	1928	561	401	231	38	1	0	0	0	0	41	201	454	(8)
	HDD65	4937	1025	817	673	360	103	11	1	2	78	342	609	916	(9)
	CDD50	3361	0	4	24	132	412	626	762	742	443	171	42	3	(10)
	CDD65	895	0	0	0	3	49	187	298	280	71	7	0	0	(11)
	CDH74	8138	0	0	3	46	415	1586	2756	2705	578	48	1	0	(12)
	CDH80	2829	0	0	0	9	73	492	1032	1090	129	4	0	0	(13)
(14) Wind	WSAvg	6.3	6.5	7.1	8.0	7.3	6.5	6.0	5.4	5.1	5.5	5.5	6.3	6.3	(14)
(15) Precipitation	PrecAvg	17.4	1.2	0.9	1.2	1.3	1.7	2.0	1.7	1.4	1.7	1.5	1.5	1.5	(15)
	PrecMax	25.5	3.8	2.4	2.9	2.6	3.9	3.7	5.6	4.6	4.9	4.1	3.6	3.9	(16)
	PrecMin	10.2	0.0	0.1	0.0	0.4	0.3	0.2	0.0	0.2	0.1	0.2	0.1	0.1	(17)
	PrecStd	3.7	0.9	0.6	0.7	0.6	1.0	1.0	1.3	1.0	1.2	1.0	1.1	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	63.2	71.6	80.2	86.4	91.6	95.2	96.1	87.9	79.2	71.2	59.4	(19)
		MCWB	48.8	52.3	53.9	62.7	67.3	72.5	71.8	72.8	66.9	64.4	59.6	52.1	(20)
	2%	DB	50.5	56.9	64.7	72.8	81.5	88.1	91.2	92.5	83.6	73.9	64.6	54.4	(21)
		MCWB	45.5	48.8	51.4	58.4	66.2	70.3	72.0	71.8	66.8	61.6	57.6	49.6	(22)
	5%	DB	46.7	52.1	60.1	68.5	78.6	85.4	88.4	89.3	80.2	70.0	60.3	50.5	(23)
		MCWB	43.4	46.0	49.8	56.3	64.4	69.6	71.3	70.5	65.7	60.6	55.2	47.0	(24)
	10%	DB	43.6	47.8	55.4	64.8	75.5	82.5	85.8	86.1	77.0	66.0	56.7	47.1	(25)
		MCWB	41.4	43.7	47.2	54.2	63.0	68.2	70.1	69.7	64.7	58.7	52.8	44.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	53.4	55.9	63.0	71.2	76.2	77.0	78.2	72.4	68.1	61.0	53.7	(27)
		MCDB	54.3	61.1	67.4	77.3	80.7	85.6	87.6	89.7	79.7	74.5	67.4	57.3	(28)
	2%	WB	46.3	49.7	52.6	60.1	68.2	73.5	75.0	75.4	70.0	64.1	58.3	50.3	(29)
		MCDB	49.7	56.0	62.7	70.2	78.2	83.4	84.6	86.2	77.7	70.2	63.7	53.6	(30)
	5%	WB	43.8	46.8	50.4	57.8	66.3	71.7	73.6	73.4	68.1	62.0	55.9	47.4	(31)
		MCDB	46.3	51.1	58.7	66.8	75.5	81.4	83.3	82.9	75.6	68.3	59.8	50.6	(32)
	10%	WB	41.4	44.2	48.1	55.7	64.3	70.0	72.1	71.7	66.3	59.7	53.6	44.5	(33)
		MCDB	43.4	47.4	54.1	63.1	72.6	79.3	82.0	82.1	74.2	64.9	56.6	46.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	13.5	16.1	18.4	19.4	19.8	20.4	21.8	20.2	17.8	14.1	12.1	(35)
		MCDBR	16.7	20.8	24.0	24.3	23.9	23.6	24.6	25.6	24.7	23.5	19.0	17.4	(36)
		MCWBR	11.8	13.6	13.3	12.5	11.1	10.2	10.3	9.8	11.3	12.5	11.6	12.5	(37)
	5% WB	MCDBR	15.2	19.4	21.9	22.2	21.5	21.0	21.9	22.8	22.1	21.3	17.5	16.5	(38)
		MCWBR	11.5	13.3	13.0	12.3	11.3	10.4	10.5	10.2	11.6	12.5	11.3	12.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.344	0.373	0.406	0.440	0.418	0.418	0.436	0.445	0.414	0.388	0.364	0.343	(40)
	taud		2.297	2.233	2.193	2.125	2.270	2.300	2.260	2.224	2.281	2.325	2.332	2.321	(41)
	Ebn at Noon		236	250	259	260	271	270	263	256	254	243	226	222	(42)
	Edh at Noon		26	34	40	46	42	41	42	42	36	30	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		387	639	1045	1518	1923	2053	2046	1833	1328	832	443	331	(44)
	RadStd		63	89	99	138	122	117	109	96	107	75	63	52	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.21	N/A	N/A	+1.60	N/A	N/A	-191	N/A	+212	+149	(46)
(47)	Regional (29 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+121	(47)

Nomenclature: See separate page